Crop Recommendation Report

Lucknow, Uttar Pradesh • 2027

Top Recommended Crop	Sugarcane	Confidence: 29.6%
District	Lucknow	Uttar Pradesh
Assessment Year	2027	Crops Evaluated: 8

Recommendation Explanation

Sugarcane is the top recommendation for Lucknow in 2027. It has consistently delivered strong yields in this district, with a historical average of 68584 kg/ha and a previous-year yield of 67242 kg/ha. Its stable cultivation history makes it a dependable choice under current climate conditions.

Full Crop Ranking

Rank	Crop	Confidence (%)	Prev. Yield (kg/ha)	Hist. Mean (kg/ha)	Area Share (%)
1	**Sugarcane**	**29.6%**	**67242**	**68584**	**23.5%**
2	Wheat	11.6%	3376	3773	9.1%
3	Rice	11.1%	3113	2984	8.9%
4	Mustard	10.7%	1099	1291	6.6%
5	Lentil	10.4%	1087	935	8.0%
6	Maize	9.7%	2999	3010	10.1%
7	Chickpea	9.6%	1175	1155	16.9%
8	Groundnut	7.3%	1611	1585	16.8%

Key Influencing Factors

Factor	Influence Direction	Magnitude
Monsoon Season Precipitation	↑ Favourable	Medium
Growing Season Temperature	→ Neutral	Low
Previous Year Yield vs Historical Average	↓ Unfavourable	Medium
District Area Share	→ Neutral	Low
Irrigation Coverage Ratio	↑ Favourable	High

Scenario Analysis

Baseline Crop	Scenario Crop	Status
Sugarcane	**Sugarcane**	**✓ Recommendation Unchanged**

Scenario Analysis Explanation

Despite minor environmental adjustments, Sugarcane retains its top position. Its strong historical yield and area share provide robustness against the simulated scenario changes.

Scenario Parameters

Parameter	Value Applied
Rainfall Change	+0%
Temperature Change	+0.0°C
Irrigation Change	+0%
Fertiliser Change	+0%

Climate Summary

Metric	Value
Annual Precipitation	990 mm
Monsoon Season Precipitation	848 mm
Average Maximum Temperature	32.0 °C
Average Minimum Temperature	18.1 °C
Irrigation Coverage Ratio	82.20%

This report was generated by the AgriCAST AI Crop Recommendation System using a LambdaMART Learning-to-Rank model trained on historical agricultural and climate data. Recommendations are decision-support tools and should be verified with local agronomists before operational use. AgriCAST v1.0